COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 / Data Structure and Analysis	**SECTION:**	CPE21S2
DATE PERFORMED:	09/02/26	**DATE SUBMITTED:**	09/02/26
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?

1. Finding an Elements Find
`for (int i = 0; i < n; i++)`
`{`
`  if (arr[i] == target)`
`  {`
`  return i;`
`  }`
`}`
	Best-Case Time Complexity : O(1)
	Worst-CaseTime Complexity : O(n)
	Brief Explanation : The target element is as the very first index of (i = 0) so that the function is going to return instant in constant time (O(1)). In the worst case, they target is at the last index or not present at all,

2.. Printing All Elements
`for (int i = 0; i < n; i++)`
`{`
`  cout << arr[i] << endl;`
`}`
	Best-Case Time Complexity : O (n)
	Worst-Case Time Complexity : O(n)
	Brief Explanation: The loop always runs from i = 0 to i = n - 1 without any conditions; breaks. It means both are finest but in the worst cases execute *n* times .

3.Nested Loop
`for (int i = 0; i < n; i++)`
`{`
`  for (int j = 0; j < n; j++)`
`  {`
`  cout << i << " " << j << endl;`
`  }`
`}`
	Best-Case Time Complexity : O(n²)
	Worst-Case Time Complexity : O(n²)
	Brief Explanation: In the outer loop it's executed n times, and for every iteration, the inner loop also execute n times.This lead to a total operation count of n X n = n^2 the input of character

4.Conditional Processing	**Best-Case Time Complexity:** O(1) **Worst-Case Time Complexity :** O(n) **Brief Explanation: (**n ≤ 100) the code it executes the constant -time else branch. As n towards infinity, it triggers the if block running loop that scales linearly with n.

``
if (n > 100)
{
    for (int i = 0; i < n; i++)
    {
        cout << arr[i];
    }
}
else
{
    cout << "Small input";
}
``	

5. Fixed Inner Loop
``
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < 10; j++)
    {
        cout << i << j;
    }
}
``
	Best-Case Time Complexity: O(n)
	Worst-Case Time Complexity :O(n)
	Brief Explanation: The n loop runs the n times.And the constant number of times is 10 the total operations counts is 10n which simplifies asymptotically to linear time.

6. Two Consecutive Loop
``
for (int i = 0; i < n; i++)
{
    cout << arr[i];
}
 	
for (int j = 0; j < n; j++)
{
    cout << arr[j] * 2;
}
``
	Best-Case Time Complexity: O(n)
	Worst-Case Time Complexity : O(n)
	Brief Explanation:

7.Diving by Two
``
int x = n;
 	
while (x > 1)
{
    x = x / 2;
}
``
	Best-Case Time Complexity:
	Worst-Case Time Complexity :
	Brief Explanation:

8.Nested Loops with Different Limit	**Best-Case Time Complexity:** **Worst-Case Time Complexity :** **Brief Explanation:**

``
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < i; j++)
    {
        cout << i << j;
    }
}
``	

9.Early Termination
``
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        cout << "Found";
        break;
    }
}
``
	Best-Case Time Complexity:
	Worst-Case Time Complexity :
	Brief Explanation:

10. Two Level Search
``
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        if (arr[i] == target)
        {
            break;
        }
    }
}
``
	Best-Case Time Complexity:
	Worst-Case Time Complexity :
	Brief Explanation:

11.Linear Loop With Logarithmic Inner Work
``
for (int i = 0; i < n; i++)
{
    int x = n;
 	
    while (x > 1)
    {
        x = x / 2;
    }
}
``
	Best-Case Time Complexity:
	Worst-Case Time Complexity :
	Brief Explanation:

12. Conditional Nested Loop	**Best-Case Time Complexity:** **Worst-Case Time Complexity :** **Brief Explanation:**

```
if (arr[0] == target)
{
    cout << "Found";
}
else
{
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
            cout << arr[i] + arr[j];
        }
    }
}
```

13. Finding the Maximum

```
int max = arr[0];

for (int i = 1; i < n; i++)
{
    if (arr[i] > max)
    {
        max = arr[i];
    }
}
```

Best-Case Time Complexity:
Worst-Case Time Complexity :
Brief Explanation:

14.Three Nested Loops

```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        for (int k = 0; k < n; k++)
        {
            cout << i + j + k;
        }
    }
}
```

Best-Case Time Complexity:
Worst-Case Time Complexity :
Brief Explanation:

15.Comprehensive Analysis
`for (int i = 0; i < n; i++)`
`{`
`  if (arr[i] == target)`
`  {`
`  return i;`
`  }`
`  int x = n;`
`  while (x > 1)`
`  {`
`  x = x / 2;`
`  }`
`}`
	Best-Case Time Complexity:
	Worst-Case Time Complexity :
	Brief Explanation:

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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